

Nutritional modulation of cognitive function and mental health.

The important role of diet in cardiometabolic health is generally well recognised; for mental health, it is not so well understood. However, lifestyle risk factors for poor physical health are the same risk factors for mental illness, including poor diet. This is reflected by the high level of poor physical health in people with mental illness. Mediterranean, whole food diets have been associated with reduced risk for chronic disease, but very little research has investigated their mental health benefits. We provide a model for the pathways by which food components provided by a Mediterranean-style diet can facilitate healthy brain function. We then review evidence for the role of selected nutrients/food components - antioxidants, omega-3 fatty acids and B vitamins - in the brain and, hence, modulation of cognitive function and mental health. Converging evidence indicates multiple pathways by which these nutrients can assist in brain function, drawing from studies investigating them in isolation. There is very little work done on synergistic actions of nutrients and whole diets, highlighting a need for human intervention studies investigating benefits of Mediterranean-style diets for mental, as well as cardiometabolic health. Copyright © 2013 Elsevier Inc. All rights reserved.